B.E. COMPUTER SCIENCE ENGINEERING

PROJECT REVIEW 2

VERA — Verified Emergency & Route Advisory

AI-assisted, context-aware and crowd-verified route safety advisory platform.

Team
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Introduction

Conventional navigation applications primarily optimize routes based on distance, travel time and traffic conditions.

However, the fastest route may not always be the safest route.

Route safety can vary depending on:

- Street lighting and road infrastructure
- Historical incident information
- Time of day and local conditions
- Public transport availability
- Road isolation and pedestrian activity
- Nearby emergency support

VERA aims to provide a safety-focused route advisory layer that evaluates route conditions alongside travel efficiency.

The Problem

Women commuting in Indian cities lack a practical, real-time way to know whether a route or stop is safe at a specific time. Navigation apps optimize for speed or distance; safety apps are reactive (SOS after an incident). Commuters need a proactive answer to: “Which route is actually safer right now, and why?”





Why This Matters

Millions of women depend on buses, metros, shared autos, and walking — often during early mornings or late evenings. Useful safety signals exist (streetlights, police posts, transit frequency) but are fragmented. Without a unified, time-sensitive view, commuters make decisions blind to real safety conditions.

- High exposure during off-peak hours
- Infrastructure data is under-utilized
- Current tools lack proactive, contextual safety guidance



Our Solution

VERA adds a dynamic safety layer on top of standard maps. Routes are color-coded (green/yellow/red) using a real-time safety score that blends infrastructure data, time-of-day adjustments, and verified crowd reports. The app helps commuters choose safer routes before they travel — proactive guidance, not just emergencies.

Objectives & approach

Objectives

- Build a safety-scoring engine for routes using streetlight density, police proximity, and transit frequency
- Design a time-of-day adjustment model — the same route scores differently at 8 AM vs 10 PM
- Build a verified, reputation-weighted crowd-reporting system with outlier detection
- Deliver a working mobile app with colour-coded routing and a one-tap SOS
- Validate the model against real Hyderabad corridors

Approach

VERA adds a route-safety scoring layer on top of standard maps — combining public infrastructure data, time-of-day risk, and verified crowd-sourced input, wrapped in a moderation pipeline that keeps the data trustworthy.

LITERATURE SURVEY

Research supports safety–efficiency trade-offs

Paper	Main focus	Relevance to VERA
Sohrabi et al. (2022), Safe route-finding: A review of literature and future directions	Reviews safe-route-finding approaches and the trade-off between safety and travel efficiency.	Supports multi-factor safety scoring and fastest-versus-safer route comparison.
Purchase & Breetzke (2026), Danger by design? Mapping safe route choices in the city of Tshwane, South Africa	Combines historical incidents, risky facilities and safety infrastructure to identify safer routes under different time conditions.	Supports context-aware, time-sensitive route assessment using environmental safety factors.

Key Observation: The safest route may differ from the shortest route, and safety conditions can vary over time.

LITERATURE SURVEY

Crime prediction and crowd-sourced route safety

Paper	Main focus	Relevance to VERA
Zhou et al. (2021), ESCORT: Fine-Grained Urban Crime Risk Inference Leveraging Heterogeneous Open Data	Estimates crime risk at individual road-segment level using spatial, temporal and historical crime information	Supports road-segment-based risk assessment and contextual safety analysis.
Zaoad et al. (2023), CrowdSPaFE: A Crowd-Sourced Multimodal Recommendation System for Urban Route Safety	Combines historical crime information and recent crowd feedback for safety-aware route recommendations.	Supports community-based safety information and safer-route recommendations.

Key Observation: Historical crime patterns and community feedback can both contribute to safety-aware route planning.

LITERATURE SURVEY

Integrated safety information and risk-aware routing

Paper	Main focus	Relevance to VERA
Mata et al. (2016), A Mobile Information System Based on Crowd-Sensed and Official Crime Data for Finding Safe Routes	Combines official crime records and crowdsourced social-media information using semantic processing and Bayesian classification to generate safer routes.	Supports combining official safety information with community reports for route assessment.
Bhumika and Das (2022), MARRS: A Framework for Multi-Objective Risk-Aware Route Recommendation Using Multitask-Transformer	Uses heterogeneous urban data and multitask learning to predict crime, accidents and traffic flow, then optimizes multiple route objectives.	Supports multi-factor risk assessment and balancing route safety with travel efficiency.

Key Observation: Combining different data sources and multiple route objectives can improve risk-aware route recommendation.

RESEARCH GAPS IDENTIFIED

01

Fragmented Data

Safety information is spread across different sources.

02

Changing Conditions

Route safety varies with time and local conditions.

03

Report Reliability

Community reports need verification and filtering.

04

Explainable Scores

Users need clear reasons behind safety ratings.

05

Data Freshness

Missing or outdated data must be identified.

06

Local Validation

Approaches need testing with Hyderabad corridor data.

VERA'S APPROACH

An integrated, explainable and context-aware route safety advisory system.

EXISTING SYSTEM

Different systems optimize different priorities

Current systems address movement or safety signals separately; an integrated route-level safe framework remains missing.

MOVEMENT

Conventional Navigation

Primarily focuses on shortest or fastest routes using distance, traffic and travel time. Safety is not the primary route-selection criterion.

SAFETY SIGNALS

Existing Safety-Oriented Platforms

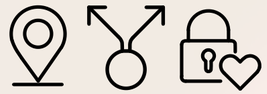
Provide safety scores, community information or emergency-support features, demonstrating the value of safety-related location information.

GAP IDENTIFIED

An integrated route-level framework is needed to combine multiple safety factors, contextual conditions, reliability-aware community observations and explainable route recommendations.

PROPOSED SYSTEM

From route request to safer action



Request & inputs

Route request — User location destination.

Alternatives — Generate alternative routes.

Safety factors — Collect and check infrastructure, incidents, activity, isolation, transport, emergency access, time/context and community reports.



Validate & compare

Reliability checkpoint — Check reliability and outliers.

Scoring engine — Calculate the safety score.

Compare — Compare routes on safety and efficiency.



Recommend & act

Explain — Explain the recommendation.

Safer route — Show the safer route and nearby emergency support.

Emergency action — Enable SOS and location sharing.

SYSTEM ARCHITECTURE

USER INTERFACE
Flutter Mobile Application
Route Request · Safety Map · SOS

↓

BACKEND SERVICES
Firebase Authentication · Cloud Functions
Request Handling · API Communication

↓

SAFETY & ROUTING ENGINE
Safety Scoring · Time Adjustment
Crowd Reliability · Route Comparison

↓

DATA LAYER
Firebase Firestore · OpenStreetMap
Infrastructure · Reports · Route Geometry

↓

ADMIN & MODERATION
React Dashboard · Report Review
Outlier Detection · Data Monitoring

What makes VERA different

Novel contributions

- Dynamic safety scoring — recalculated by time of day, not a static label
- Blended trust model — objective data + reputation-weighted crowd input
- Outlier & fraud detection — spam bursts quarantined before reaching live scores
- Privacy-first — no continuous tracking, location captured only at report/SOS moments

Technology stack

Layer	Tool
Mobile App	Flutter
Backend / DB	Firebase (Firestore, Auth, Functions)
Maps & Routing	Google Maps / OpenStreetMap
Scoring Engine	Node.js (weighted scoring model)
Moderation Dashboard	React

IMPLEMENTATION PROGRESS

COMPLETED

- Safety scoring modules implemented, including time-based adjustments and crowd-report reliability checks.
- Route geometry imported from OpenStreetMap for the Tolichowki test corridor.
- 618 road segments successfully parsed and imported into the local Firestore emulator.
- React-based dashboard map updated to display road-segment geometry.
- Dashboard production build completed successfully.

IN PROGRESS

- Integrating streetlight, public transport and other infrastructure information.
- Improving route safety score explanations and data reliability.
- Verifying map visualization and end-to-end route assessment.

CURRENT LIMITATIONS

- Some infrastructure fields, including streetlight density and transit waiting time, are currently missing.
- Imported road geometry alone is not sufficient to produce meaningful safety scores.
- Real-world corridor validation and live data integration are still pending.

FUNCTIONAL TESTING

- Safety scoring logic and time-of-day adjustments.
- Reputation-weighted crowd-report processing.
- Outlier detection and report reliability checks.
- Route comparison and recommendation logic.

DATA VALIDATION

- OpenStreetMap road-segment data imported for the Tolichowki test corridor.
- Road geometry checked for missing or invalid coordinates.
- Test data kept separate from existing synthetic test records using a separate corridor identifier.

VALIDATION PENDING

- Testing with complete streetlight and public transport data.
- End-to-end verification of the dashboard map and route recommendations.
- Evaluation using real-world Hyderabad corridor conditions.
- Validation of data freshness and safety-score reliability.

Methodology



The workflow combines multiple route safety factors, contextual conditions and reliability-checked community observations to calculate an explainable safety score and recommend a safer route.

FUTURE SCOPE

01

Live Data Integration

Integrate updated infrastructure, public transport and relevant safety information.

02

Advanced Safety Analysis

Improve risk assessment using additional contextual and environmental factors.

03

Enhanced Crowd Verification

Strengthen report verification, reputation scoring and suspicious activity detection.

04

Real-Time Route Updates

Recalculate route scores when new information becomes available.

05

Expanded Geographic Coverage

Extend testing from selected Hyderabad corridors to additional areas.

06

Improved Emergency Support

Enhance SOS functionality, location sharing and access to nearby emergency services.

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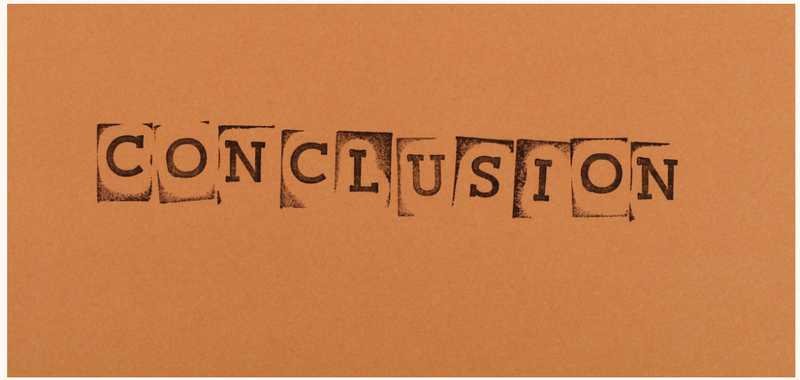
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- VERA proposes a safety-focused route advisory system that goes beyond conventional distance- and time-based navigation.
- It combines infrastructure information, time-based conditions and community reports to assess route safety.
- Its scoring and route comparison approach aims to help users make more informed travel decisions.
- The current prototype demonstrates initial progress in safety scoring, OpenStreetMap road data integration and dashboard development.
- Further data integration, testing and real-world validation are required before practical deployment.

VERA — Towards safer, more informed journeys.